

How Do We Prove Transition Integrity in Agentic AI?

Not T₄. Not T₅. The Space Between Them.

A trustworthy AI decision does not automatically prove trustworthy execution.

The assurance gap is addressed by the

Agentic AI T₄ → T₅ Assurance Cone Framework™ V3.0

Transition Integrity | Execution-Path Coverage | Boundary Integrity | Independent Assurance

SIX ASSURANCE FUNCTIONS.

ONE ACCOUNTABLE CONCLUSION. (I-A-N-E-T-F)

- 1 INTENT**
What was intended and why does it matter?
 - Consequential objective
 - Affected resource
 - Materiality
 - Permitted / prohibited purpose
 - Applicable constraints
- 2 AUTHORITY**
Is the transition authorized?
 - Who/what can authorize
 - Scope and duration
 - Resource and action
 - Authority state at execution point
 - Delegations and constraints
- 3 ENFORCEMENT**
What actually controlled execution?
 - Policy vs actual enforcement
 - Permit / deny / contain / terminate
 - Control effectiveness
 - No bypass paths
 - Enforcement at commit point
- 4 TESTING**
Can the conclusion be independently challenged?
 - Adversarial testing
 - Stale or revoked authority
 - Parallel / asynchronous paths
 - Replay / duplication / bypass
 - Execution-path coverage
 - Failure and recovery scenarios
- 5 TRACEABILITY & EVIDENCE**
Can the transition be reconstructed?
 - Who → What → Why
 - Authority → Policy → Path
 - Enforcement → Commit™
 - Downstream lineage
 - Evidence integrity and provenance
- 6 FORENSICS & REVOCABILITY**
Can the transition be traced and contained?
 - Event reconstruction
 - Exposure and containment
 - Revocation effectiveness
 - Residual effects
 - Accountability and responsibility

ENTERPRISE CONTEXT
RISK | PEOPLE | PROCESSES | TECHNOLOGY | DATA | VALUE



THE T₄ → T₅ ASSURANCE BOUNDARY (V3.0)

Must be established, within the defined assurance scope:

- AUTHORITY VALIDATION**
Was the relevant authority valid at the applicable commit point?
- TRANSITION CONTROL**
Were required controls active and effective?
- EXECUTION-PATH COVERAGE**
Were all consequential execution paths identified, controlled, and accounted for?
- INTERVENTION / REVOCATION EFFECTIVENESS**
If intervention occurred, did it become operationally effective?
- EVIDENCE RECONCILIATION**
Do independent evidence sources establish a consistent transition sequence?
- BOUNDARY INTEGRITY**
Does the assurance conclusion remain within the demonstrated evidence and execution boundary?

NO ASSURANCE CLAIM MAY EXTEND BEYOND WHAT CAN BE INDEPENDENTLY DEMONSTRATED.

KEY V3.0 PRINCIPLES

- AUTHORITY ≠ CAPABILITY
- AUTHORIZATION ≠ EXECUTION
- EVIDENCE VALIDITY ≠ EVIDENCE COMPLETENESS
- COVERAGE MUST BE DEMONSTRATED
- UNKNOWN COVERAGE IS A LIMITATION
- NO CAPTURED ACT ≠ NO ACT OCCURRED
- ASSURANCE ≠ ASSERTION
- CONCLUSION CANNOT EXCEED EVIDENCE
- ALIGNMENT ≠ EXTERNAL CONFORMITY
- FRAMEWORK USE ≠ CERTIFICATION

THE GOVERNING ASSURANCE QUESTION (V3.0)

Can an independent examiner establish — within the defined assurance scope — that the consequential transition from T₄ to T₅ occurred through an authorized and controlled execution path, and that no unaccounted consequential execution path exists capable of producing the relevant effect within that scope?

YES

Transition Integrity Demonstrated
Accept the conclusion

WITH LIMITATIONS

Partially Demonstrated
Qualify the conclusion

NO

Transition Integrity Unproven
Reject the conclusion

THE T₄ → T₅ LIFECYCLE (V3.0)



Adnan Shamim

Founder | The First Agentic AI Audit Ecosystem™



RESPONSIBLE AI



INDEPENDENT ASSURANCE



REAL-WORLD IMPACT

TRUST
BUILDS
TOMORROW

THE AGENTIC AI T₄ → T₅ ASSURANCE CONE FRAMEWORK™ V3.0
Independent Assurance for a Safer, More Accountable AI Future.

THE AGENTIC AI $T_4 \rightarrow T_5$ ASSURANCE CONE FRAMEWORK™

V3.0 — Enterprise Assurance Edition

Transition Integrity • Execution-Path Coverage • Boundary Integrity • Independent Assurance

Not T_4 . Not T_5 . The Space Between Them.

1. PURPOSE worthy AI decision does not automatically prove trustworthy execution.

The Agentic AI $T_4 \rightarrow T_5$ Assurance Cone Framework™ establishes a technology-neutral independent-assurance structure for evaluating whether a consequential transition from an observed lifecycle state through commit-time execution to an authoritative enforcement and downstream state is sufficiently governed, attributable, controlled, evidenced, reconstructable, and bounded.

The Framework is independently structured and does not prescribe a particular technology, architecture, vendor, protocol, implementation pattern, or control product.

2. FRAMEWORK STATUS & NON-SUBSTITUTION

The Agentic AI $T_4 \rightarrow T_5$ Assurance Cone Framework™ is an independently developed assurance framework.

It is not intended to:

- replace applicable laws or regulations;
- replace organizational governance requirements;
- replace applicable risk-management requirements;
- replace information-security requirements;

- replace internal-control requirements;

- replace internal-audit professional requirements;

- establish regulatory approval;

- establish certification or accreditation;

- automatically establish compliance with any external requirement.

The Framework may be mapped to, used alongside, or incorporated within an organization's existing governance, risk, compliance, security, audit, and assurance architecture.

Application of this Framework does not, by itself, establish conformity with any external requirement.

3. EXTERNAL REQUIREMENTS & ALIGNMENT

Where an organization is subject to external laws, regulations, contractual obligations, professional requirements, governance requirements, or other control obligations, those requirements remain independently applicable.

V3.0 may be used as an assurance overlay to evaluate the $T_4 \rightarrow T_5$ transition against those applicable requirements.

The organization must separately determine:

- which external requirements apply;
- which requirements are mandatory;
- which controls are required;
- which evidence is required;
- which conformity criteria apply;
- which reporting obligations exist;
- which regulatory or contractual consequences apply.

Governing Principle

Framework alignment does not establish external conformity.

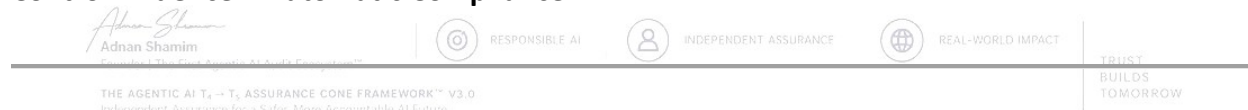
Therefore:

V3.0 Application $\neq$ External Compliance

Framework Alignment $\neq$ Certification

Assurance Conclusion $\neq$ Regulatory Approval

Control Evidence $\neq$ Automatic Compliance



4. CLAIMS DISCIPLINE

Any assurance statement produced using V3.0 must clearly identify, where material:

1. the assurance population;
2. the relevant time period;
3. the execution-path boundary;
4. the evidence boundary;
5. material exclusions;
6. material limitations;
7. the assurance classification;
8. the applicable control requirements;
9. the basis for the conclusion.

No statement may imply a broader assurance conclusion than the underlying evidence, coverage, authority, enforcement, or defined boundary supports.

No V3.0 conclusion may be represented as certification, accreditation, regulatory approval, or automatic conformity with an external requirement.

5. FUNDAMENTAL ASSURANCE PROBLEM

A trustworthy AI decision does not automatically establish trustworthy execution.

Likewise:

Observed State ≠ Authoritative State

Capability ≠ Authority

Authority ≠ Authorization

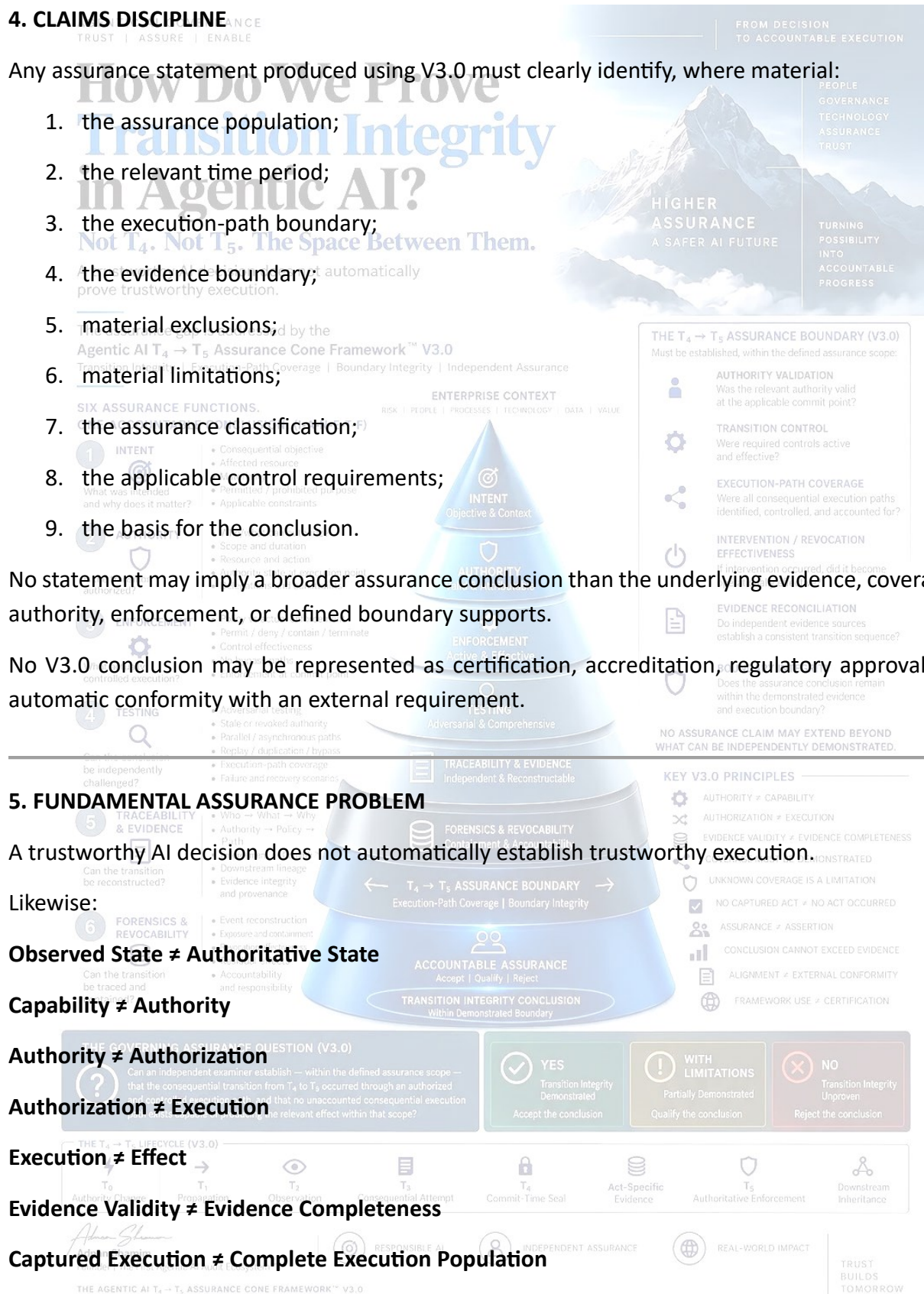
Authorization ≠ Execution

Execution ≠ Effect

Evidence Validity ≠ Evidence Completeness

Captured Execution ≠ Complete Execution Population

Propagation Completion ≠ Retroactive Authorization



T₅ Enforcement ≠ Retroactive Validation of T₄

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Therefore, trustworthy endpoints alone cannot establish transition integrity.

6. THE T₄→T₅ ASSURANCE SEAM

T₄ — OBSERVATION & COMMIT-TIME SEALING

T₄ establishes what the actor observed and what transition context was sealed, including where applicable:

- lifecycle state;
- freshness;
- propagation context;
- execution path;
- attempted transition;
- relevant conditions;
- commit-time context.

T₄ observation does not, by itself, establish authoritative authorization.

ACT-SPECIFIC COMMIT EVIDENCE

A consequential act requires evidence capable of binding:

ACT + ACTOR + AUTHORITY + POLICY + PATH + ENFORCEMENT + TIME + COMMIT + EFFECT + LINEAGE

T₅ — AUTHORITATIVE ENFORCEMENT & DOWNSTREAM INHERITANCE

T₅ establishes what the authoritative enforcement environment actually permitted, denied, constrained, contained, terminated, or otherwise governed.

T₅ must not be used as retroactive authorization for an earlier consequential act.

7. T₀–T₅ ADVERSARIAL TRACE

T₀ — AUTHORITY CHANGE

Revocation, suspension, restriction, expiry, policy change, or other material authority event.

T₁ — PROPAGATION

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Changed authority state propagates through relevant systems, agents, services, queues, credentials, caches, delegated processes, and execution paths.

T₂ — OBSERVATION

Actor/agent observes the available lifecycle or authority state.

T₃ — CONSEQUENTIAL ATTEMPT

Actor/agent initiates or attempts a consequential action.
The assurance gap is addressed by the

T₄ — COMMIT-TIME OBSERVATION & SEAL

Observed state, freshness, execution path, and transition context are preserved.

ACT-SPECIFIC EVIDENCE

Evidence binds the consequential act to the applicable authority and enforcement conditions.

T₅ — AUTHORITATIVE ENFORCEMENT

Authoritative control determines the downstream treatment of the transition.

8. I-A-N-E-T-F ASSURANCE CONE™

I — INTENT

Establish:

- intended objective;
- consequential action;
- affected resource;
- materiality;
- permitted purpose;
- prohibited purpose;
- applicable constraints.

A — AUTHORITY

Establish:

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THE T₄ → T₅ ASSURANCE BOUNDARY (V3.0)

Must be established, within the defined assurance scope:

- AUTHORITY VALIDATION**
Was the relevant authority valid at the applicable commit point?
- TRANSITION CONTROL**
Were required controls active and effective?
- EXECUTION-PATH COVERAGE**
Were all consequential execution paths?
- INTERVENTION / REVOCATION EFFECTIVENESS**
If intervention occurred, did it become operationally effective?
- EVIDENCE RECONCILIATION**
Do the relevant evidence sources establish a consistent transition sequence?
- BOUNDARY INTEGRITY**
Does the assurance conclusion remain within the demonstrated evidence and execution boundary?

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THE GOVERNING ASSURANCE QUESTION (V3.0)

- Establish — within the defined assurance scope — that the consequential transition from T₄ to T₅ occurred through an authorized and controlled execution path, and that no unaccounted consequential execution exists or is probable of producing the relevant effect within that scope?

YES
Transition Integrity Demonstrated
Accept the conclusion

WITH LIMITATIONS
Partially Demonstrated
Qualify the conclusion

NO
Transition Integrity Unproven
Reject the conclusion

THE T₄ → T₅ LIFECYCLE (V3.0)



Adnan Shamim

Author | The First Agentic AI Audit Ecosystem™

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- who/what could authorize;

How Do We Prove Transition Integrity in Agentic AI?

- scope;
- duration;
- resource;
- action;
- applicable conditions;
- authority state at the relevant execution boundary.

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N — ENFORCEMENT

Establish what actually controlled execution.

Distinguish:

Policy Exists

from

Policy Was Enforced

from

Enforcement Affected Execution

E — TESTING

Test whether the assurance claim survives adversarial conditions, including:

- revoked authority;
- stale authority;
- propagation delay;

policy drift;

failed enforcement;

bypass;

replay;

duplication;

parallel execution;



SIX ASSURANCE FUNCTIONS.

1 INTENT

- What was intended and why does it matter?
- Consequential objective
- Affected resource
- Materiality
- Permitted / prohibited purpose
- Applicable constraints

2 AUTHORITY

- Who/what can authorize
- Scope and duration
- Resource and action
- Authority state at execution point
- Delegations and constraints

3 ENFORCEMENT

- Policy vs actual enforcement
- Permit / deny / contain / terminate
- Control effectiveness
- No bypass paths
- Enforcement at commit point

4 TESTING

- Adversarial testing
- Can the conclusion be independently demonstrated?
- Policy / enforcement / authority
- Replay / duplication / bypass
- Execution-path coverage
- Failure and recovery scenarios

5 TRACEABILITY

- Who → What → Why
- Can the transition be independently demonstrated?
- Enforcement → Commit
- Downstream lineage
- Containment & Accountability

6 FORENSICS & RECOVERY

- Event reconstruction
- Can the transition be independently demonstrated?
- Containment & Accountability
- Event reconstruction
- Containment & Accountability
- Residual effects
- Accountability
- Containment & Accountability

ENTERPRISE CONTEXT

RISK | PEOPLE | PROCESSES | TECHNOLOGY | DATA | VALUE



THE $T_4 \rightarrow T_5$ ASSURANCE BOUNDARY (V3.0)

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THE GOVERNING ASSURANCE QUESTION (V3.0)

Can an independent examiner establish — within the defined assurance scope — that no T_4 to T_5 occurred through an authorized and controlled execution path such that no unaccounted consequential execution path exists capable of producing the relevant effect within that scope?

YES
Transition Integrity Demonstrated
Accept the conclusion

WITH LIMITATIONS
Partially Demonstrated
Qualify the conclusion

NO
Transition Integrity Unproven
Reject the conclusion

THE ASSURANCE CYCLE (V3.0)



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THE FIRST AGENTIC AI AUDIT ECOSYSTEM™ V3.0
The first framework for a safer, more accountable AI future.

- asynchronous execution;

- delegated execution;

- cached authorization;

- alternate credentials;

- failover;

- recovery; AI decision does not automatically prove trustworthy execution.

- compromised evidence generation.

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Execution-Path Coverage is a mandatory Testing dimension.

T — TRACEABILITY & EVIDENCE

Establish:

Who → What → Why → Authority → Policy → Path → Enforcement → Commit → Effect → Downstream Lineage → Evidence

F — FORENSICS & REVOCABILITY

Establish the ability to:

- reconstruct events;
- identify authority;
- identify execution path;
- establish exposure;
- establish containment;
- demonstrate revocation effectiveness;
- identify residual effects;
- preserve evidence;
- determine responsibility;

- identify where the assurance boundary was exceeded.



SIX ASSURANCE FUNCTIONS.

ENTERPRISE CONTEXT

RISK | PEOPLE | PROCESSES | TECHNOLOGY | DATA | VALUE

1 INTENT



What was intended and why does it matter?

- Consequential objective
- Affected resource
- Materiality
- Permitted / prohibited purpose
- Applicable constraints

2 AUTHORITY



Is the transition authorized?

- Scope and duration
- Delegations and constraints

3 ENFORCEMENT



Was the transition controlled?

- Permit / deny / contain / terminate
- Control effectiveness

4 TESTING



Can the execution be challenged?

- Bypass paths
- Enforcement at commit point

5 TRACEABILITY & EVIDENCE



Can the execution be reconstructed?

- Parallel / asynchronous paths
- Replay / duplication / bypass
- Path coverage
- Path recovery scenarios

6 FORENSICS & REVOCABILITY



Can the execution be traced and contained?

- Authority
- Path
- Enforcement → Commit
- Lineage
- Integrity and provenance



THE $T_4 \rightarrow T_5$ ASSURANCE BOUNDARY (V3.0)

Must be established, within the defined assurance scope:



AUTHORITY VALIDATION

Was the relevant authority valid at the applicable commit point?



TRANSITION CONTROL

Were required controls active and effective?



EXECUTION-PATH COVERAGE

Were all consequential execution paths identified, controlled, and accounted for?



INTERVENTION / REVOCATION EFFECTIVENESS

If intervention occurred, did it become operationally effective?



EVIDENCE RECONCILIATION

Do independent evidence sources establish a consistent transition sequence?



BOUNDARY INTEGRITY

Does the assurance conclusion remain within the demonstrated evidence and execution boundary?

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KEY V3.0 PRINCIPLES

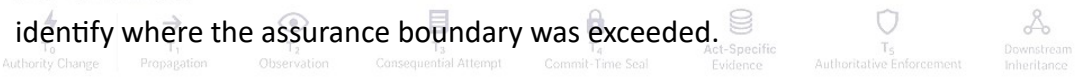
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- ALIGNMENT $\neq$ EXTERNAL CONFORMITY
- FRAMEWORK USE $\neq$ CERTIFICATION

THE GOVERNING ASSURANCE QUESTION (V3.0)

- Establish — within the defined assurance scope — that the consequential transition from T_4 to T_5 occurred through an authorized and controlled execution path, and that no unaccounted consequential execution path exists capable of producing a resultant effect within that scope?
- determine responsibility;

YES	WITH LIMITATIONS	NO
Transition Integrity Demonstrated	Partially Demonstrated	Transition Integrity Unproven
Accept the conclusion	Qualify the conclusion	Reject the conclusion

THE $T_4 \rightarrow T_5$ LIFECYCLE (V3.0)



9. EXECUTION-PATH COVERAGE

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Execution-Path Coverage is a mandatory assurance condition.

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The claimed consequential execution population must be defined.

Potential paths include:

- primary execution;
- parallel execution;
- asynchronous queues;
- scheduled execution;
- delegated agents;

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Transition Integrity | Execution-Path Coverage | Boundary Integrity | Independent Assurance

SIX ASSURANCE FUNCTIONS.

- 1 INTENT
 - Consequential objective
 - Affected resource
 - Materiality
 - Permitted / prohibited purpose
 - Applicable constraints
- 2 AUTHORITY
 - Scope and duration
 - Resource and action
- 3 ENFORCEMENT
 - Policy vs actual enforcement
 - Control effectiveness
 - No bypass paths
- 4 TESTING
 - Adversarial testing
 - Scale or limited authority
 - Replay / duplication / bypass
 - Execution-path coverage
 - Failure and recovery scenarios
- 5 TRACEABILITY
 - Who → What → Why
 - Authority → Policy → Path
 - Enforcement → Commit™
 - Downstream lineage
- 6 FORENSICS & REVOCABILITY
 - Event reconstruction
 - Reasonableness
 - Residual effects
 - Accountability

ENTERPRISE CONTEXT
RISK | PEOPLE | PROCESSES | TECHNOLOGY | DATA | VALUE



THE $T_4 \rightarrow T_5$ ASSURANCE BOUNDARY (V3.0)

Must be established, within the defined assurance scope:

- AUTHORITY VALIDATION
 - Was the relevant authority valid at the applicable commit point?
- TRANSITION CONTROL
 - Were required controls active and effective?
- EXECUTION-PATH COVERAGE
 - Were all consequential execution paths identified, controlled, and accounted for?
- INTERVENTION / REVOCATION EFFECTIVENESS
 - If intervention occurred, did it become operationally effective?
- EVIDENCE RECONCILIATION
 - Do independent evidence sources establish a consistent transition sequence?
- BOUNDARY INTEGRITY
 - Does the assurance conclusion remain within the demonstrated evidence and execution boundary?

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A path capable of producing the claimed consequence cannot be silently assumed to be covered.

GOVERNING ASSURANCE QUESTION (V3.0)			
Can an independent examiner establish — within the defined assurance scope — that the consequential transition from T_4 to T_5 occurred through an authorized and controlled execution path, and that no unaccounted consequential execution path exists capable of producing the relevant effect within that scope?	YES Transition Integrity Demonstrated	WITH LIMITATIONS Partially Demonstrated	NO Transition Integrity Unproven
	Accept the conclusion	Qualify the conclusion	Reject the conclusion

10. DEMONSTRABLE COVERAGE CRITERIA

Execution-path coverage is **Demonstrated** only where applicable evidence establishes:

C1 — POPULATION DEFINITION

The consequential execution population is explicitly identified.

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C2 — PATH IDENTIFICATION

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Known paths capable of producing the consequence are identified.

C3 — CONTROL BOUNDARY

Each claimed covered path is subject to the required authority and enforcement boundary.

C4 — OBSERVABILITY

Not T₄. Not T₅. The Space Between Them.

Relevant consequential activity is sufficiently observable to support the specific assurance claim.

C5 — EVIDENCE BINDING

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Consequential acts are attributable to the relevant execution path and commit context.

C6 — BYPASS ASSESSMENT

SIX ASSURANCE FUNCTIONS. (I-A-N-E-T-F)

Known alternate, parallel, asynchronous, delegated, recovery, and failover paths are addressed.

C7 — EXCLUSION DISCLOSURE

Any path outside coverage is explicitly identified.

C8 — INDEPENDENT RECONSTRUCTION

An examiner can independently reconstruct the relevant transition and determine the applicable assurance boundary.

C9 — FAILURE CONSEQUENCE

The predetermined assurance consequence of insufficient coverage is defined.

Coverage is not demonstrated merely because the primary path is instrumented.

11. COVERAGE CLASSIFICATION

Every relevant execution path should be classified as:

COVERED

Can an independent examiner establish — within the defined assurance scope — that the consequential transition from T₄ to T₅ occurred through an authorized and controlled execution path, and that no unaccounted consequential execution path exists capable of producing the relevant effect within that scope?

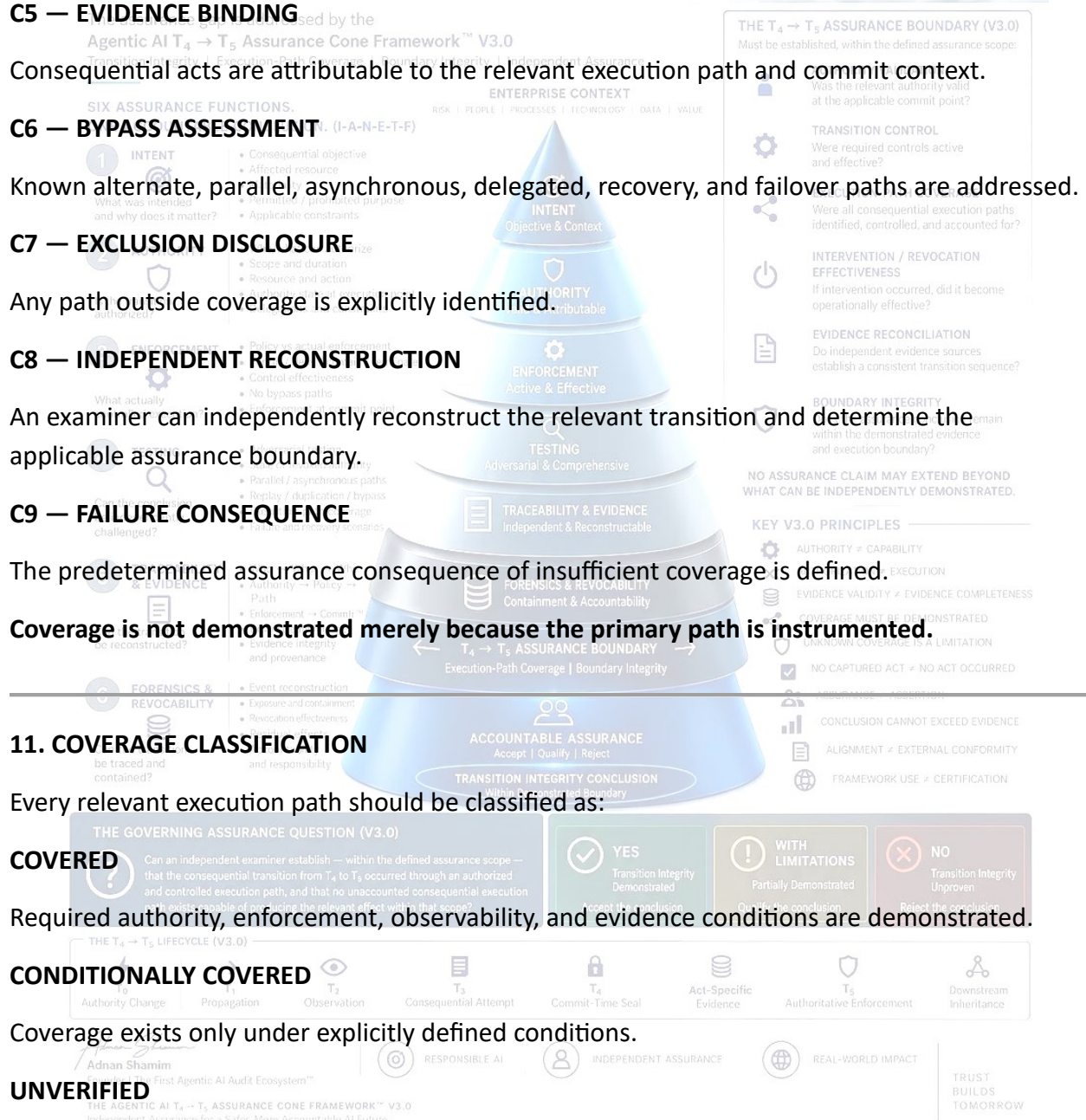
Required authority, enforcement, observability, and evidence conditions are demonstrated.

CONDITIONALLY COVERED

Coverage exists only under explicitly defined conditions.

UNVERIFIED

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Coverage cannot be independently demonstrated.

EXCLUDED

The path is deliberately outside the assurance boundary.

FAILED

The path was demonstrated to violate a required assurance condition.

A trustworthy AI decision does not automatically prove trustworthy execution.

12. EXECUTION-PATH COVERAGE INVARIANT

Agentic AI $T_4 \rightarrow T_5$ Assurance Cone Framework™ V3.0

An assurance conclusion shall not extend beyond the consequential execution population and execution paths for which the required authorization, enforcement, detection, and evidence boundary can be independently demonstrated.

Therefore:

Covered → Assurance Permitted

Conditionally Covered → Assurance Qualified

Unverified → Assurance Limited

Excluded → Outside Conclusion

Failed → Control/Assurance Failure

13. BOUNDARY INTEGRITY

The assurance boundary must explicitly state:

- what is included;
- what is excluded;

- what is controlled;

- what is observable;

- what is evidenced;

- what is reconstructable;

- what remains unknown;

- what consequence follows from the limitation.

TRUST | ASSURE | ENABLE

FROM DECISION
TO ACCOUNTABLE EXECUTION

PEOPLE
GOVERNANCE
TECHNOLOGY
ASSURANCE
TRUST

HIGHER
RANGE
A SAFER AI FUTURE

TURNING
POSSIBILITY
INTO
ACCOUNTABLE
PROGRESS

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TRANSITION INTEGRITY
Transition Integrity | Execution-Path Coverage | Boundary Integrity | Independent Assurance

ONE ACCOUNTABLE CONCLUSION. (I-A-N-E-T-I-P)

1 AUTHORITY
What is intended and why does it matter?
• Affected resource
• Materiality
• Permitted / prohibited purpose
• Applicable constraints

2 ENFORCEMENT
What actually controlled execution?
• Enforcement at commit point

3 TESTING
Can the transition be independently challenged?
• State or revoked authority
• Realistic / asynchronous paths
• Execution-path coverage
• Failure and recovery scenarios

4 TRACEABILITY & EVIDENCE
Who → What → Why
• Authority → Policy → Commit
• Downstream lineage
• Evidence integrity and provenance

5 FORENSICS & REVOCABILITY
Can the transition be traced and reconstructed?
• Event reconstruction
• Exposure and containment
• Effectiveness
• Accountability and responsibility

6 ACCOUNTABLE ASSURANCE
Accept | Qualify | Reject

TRANSITION INTEGRITY CONCLUSION
Within Demonstrated Boundary

ENTERPRISE CONTEXT

INTENT
Objective & Context

AUTHORITY
Valid & Attributable

ENFORCEMENT
Active & Effective

TESTING
Adversarial & Comprehensive

TRACEABILITY & EVIDENCE
Independent & Reconstructable

FORENSICS & REVOCABILITY
Containment & Accountability

ACCOUNTABLE ASSURANCE
Accept | Qualify | Reject

TRANSITION INTEGRITY CONCLUSION
Within Demonstrated Boundary

THE T₄ → T₅ ASSURANCE BOUNDARY (V3.0)
Must be established, within the defined assurance scope:

1. Was the relevant authority valid and attributable to the commitment?

2. Were required controls active and effective?

3. Were all consequential execution paths identified, controlled, and accounted for?

4. If intervention occurred, did it become operationally effective?

5. Do independent evidence sources establish a consistent transition sequence?

6. Does the assurance conclusion remain within the demonstrated evidence and execution boundary?

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NO CAPTURED ACT ≠ NO ACT OCCURRED

ASSURANCE ≠ ASSERTION

CONCLUSION CANNOT EXCEED EVIDENCE

ALIGNMENT ≠ EXTERNAL CONFORMITY

FRAMEWORK USE ≠ CERTIFICATION

THE T₄ → T₅ LIFECYCLE (V3.0)

T₀ Authority Change

T₁ Propagation

T₂ Observation

T₃ Consequential Attempt

T₄ Commit-Time Seal

Act-Specific Evidence

Authoritative Enforcement

Downstream Inheritance

RESPONSIBLE AI

INDEPENDENT ASSURANCE

REAL-WORLD IMPACT

TRUST BUILDS TOMORROW

Boundary Integrity Rule

TRUST | ASSURE | ENABLE

Unknown coverage is an assurance condition—not evidence of absence.

Therefore:

No Captured Act ≠ No Act Occurred

No Observed Violation ≠ Demonstrated Prevention

A trustworthy AI decision does not automatically prove trustworthy execution.

14. PROPAGATION-WINDOW ASSURANCE

Agentic AI $T_4 \rightarrow T_5$ Assurance Cone Framework™ V3.0

Distributed-system propagation latency is not automatically a control failure.

However, consequential exposure during the propagation window requires predefined treatment.

The assurance boundary should establish, where applicable:

- maximum exposure duration;
- affected authority scope;
- permitted actions;
- prohibited actions;
- enforcement controls;
- containment;
- compensating controls;
- monitoring;
- escalation;
- accountability;
- residual-risk ownership;
- boundary-exceedance criteria;



Critical distinction

Measuring Exposure ≠ Proving Prevention

Independent Assurance for a Safer, More Accountable AI Future.



SIX ASSURANCE FUNCTIONS.

1. INTENT
 - Consequential objective
 - Affected resource
 - Materiality
 - Permitted / prohibited purpose
 - Applicable constraints
2. AUTHORITY
 - Who/what can authorize
 - Scope and duration
 - Delegations and constraints
3. ENFORCEMENT
 - Permit / deny / contain / terminate
 - Control effectiveness
 - Paths controlled execution?
4. TESTING
 - Adversarial / comprehensive
 - Parallel / asynchronous paths
 - Replay / duplication / bypass
 - Failure and recovery scenarios
5. TRACEABILITY & EVIDENCE
 - Independent & Reconstructable
 - Who → What → Why
 - Authority → Policy → Path
 - Enforcement → Commit
 - Accountability and provenance
6. FORENSICS & REVOCABILITY
 - Containment & Accountability
 - Event reconstruction
 - Exposure and containment
 - Revocation effectiveness
 - Residual effects
 - Accountability and responsibility

ENTERPRISE CONTEXT

RISK | PEOPLE | PROCESSES | TECHNOLOGY | DATA | VALUE



THE $T_4 \rightarrow T_5$ ASSURANCE BOUNDARY (V3.0)

Must be established, within the defined assurance scope:

- AUTHORITY VALIDATION
 - Was the relevant authority valid at the applicable commit point?
- ENFORCEMENT VALIDATION
 - Were required controls active and effective?
- EXECUTION-PATH COVERAGE
 - Were all consequential execution paths identified, controlled, and accounted for?
- INTERVENTION / REVOCATION EFFECTIVENESS
 - If intervention occurred, did it become operationally effective?
- EVIDENCE RECONCILIATION
 - Do independent evidence sources establish a consistent transition sequence?
- BOUNDARY INTEGRITY
 - Does the assurance conclusion remain within the demonstrated evidence and execution boundary?

NO ASSURANCE CLAIM MAY EXTEND BEYOND WHAT CAN BE INDEPENDENTLY DEMONSTRATED.

KEY V3.0 PRINCIPLES

- AUTHORITY ≠ CAPABILITY
- AUTHORIZATION ≠ EXECUTION
- EVIDENCE VALIDITY ≠ EVIDENCE COMPLETENESS
- COVERAGE MUST BE DEMONSTRATED
- UNKNOWN COVERAGE IS A LIMITATION
- NO CAPTURED ACT ≠ NO ACT OCCURRED
- ASSURANCE ≠ ASSERTION
- CONCLUSION CANNOT EXCEED EVIDENCE
- ALIGNMENT ≠ EXTERNAL CONFORMITY
- FRAMEWORK USE ≠ CERTIFICATION

THE GOVERNING ASSURANCE QUESTION (V3.0)

- Within the defined assurance scope — that the consequential transition from $T_4 \rightarrow T_5$ occurred through an authorized and controlled execution path, and that no unaccounted consequential execution path exists capable of producing the relevant effect within that scope?



AGENTIC AI GOVERNANCE

TRUST | ASSURE | ENABLE

FROM DECISION

TO ACCOUNTABLE EXECUTION

15. MULTI-AGENT & DELEGATED AUTHORITY

Where authority moves between agents, systems, or services, sufficient lineage must establish:

Origin → Delegation → Authority → Policy → Execution Path → Enforcement → Commit → Effect → Downstream Inheritance

Delegation must not silently expand effective authority.

How Do We Prove Transition Integrity in Agentic AI?

Not T₄. Not T₅. The Space Between Them.

A trustworthy AI decision does not automatically prove trustworthy execution.

PEOPLE GOVERNANCE

TRUST

HIGHER ASSURANCE

A SAFER AI FUTURE

TURNING POSSIBILITY INTO ACCOUNTABLE PROGRESS

The assurance gap is addressed by the

Agentic AI T₄ → T₅ Assurance Cone Framework™ V3.0

Addressing the Assurance Gap Between Boundary Integrity | Independent Assurance

16. ALTERNATE-PATH ASSURANCE

Any alternate path capable of producing the claimed consequence must be:

Controlled

or

Evidenced

or

Detected

or

Explicitly Excluded

or

Treated as an Assurance Limitation

ENTERPRISE CONTEXT

INTENT

Objective & Context

AUTHORITY

Valid & Attributable

ENFORCEMENT

Active & Effective

TESTING

Adversarial & Comprehensive

TRACEABILITY & EVIDENCE

Independent & Reconstructable

FORENSICS & REVOCABILITY

Containment & Accountability

TRANSITION INTEGRITY CONCLUSION

Accept | Quality | Reject

THE T₄ → T₅ ASSURANCE BOUNDARY (V3.0)

Must be established, within the defined assurance scope:

AUTHORITY VALIDATION

Was the relevant authority valid at the applicable commit point?

TRANSITION CONTROL

Were required controls active and effective?

EXECUTION-PATH COVERAGE

Were all consequential execution paths identified, controlled, and accounted for?

INTERVENTION / REVOCATION EFFECTIVENESS

If intervention occurred, did it become operationally effective?

EVIDENCE RECONCILIATION

Do independent evidence sources establish a consistent transition sequence?

BOUNDARY INTEGRITY

Does the assurance conclusion remain within the demonstrated evidence and execution boundary?

NO ASSURANCE CLAIM MAY EXTEND BEYOND WHAT CAN BE INDEPENDENTLY DEMONSTRATED.

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FRAMEWORK USE ≠ CERTIFICATION

17. ACT-SPECIFIC EVIDENCE REQUIREMENT

A consequential act should have evidence capable of binding:

ACT + ACTOR + AUTHORITY + POLICY + PATH + ENFORCEMENT + TIME + COMMIT + EFFECT + LINEAGE

THE GOVERNING ASSURANCE QUESTION (V3.0)

that the consequential transition from T₄ to T₅ occurred through an authorized and controlled execution path, and that no unaccounted consequential execution path exists capable of producing the relevant effect within that scope?

Transition Integrity Demonstrated

Accept the conclusion

Transition Integrity Partially Demonstrated

Qualify the conclusion

Transition Integrity Unproven

Reject the conclusion

Evidence should be:

THE T₄ → T₅ LIFECYCLE (V3.0)

T₀ Authority Change

T₁ Propagation

T₂ Observation

T₃ Consequential Attempt

T₄ Commit-Time Seal

Act-Specific Evidence

T₅ Authoritative Enforcement

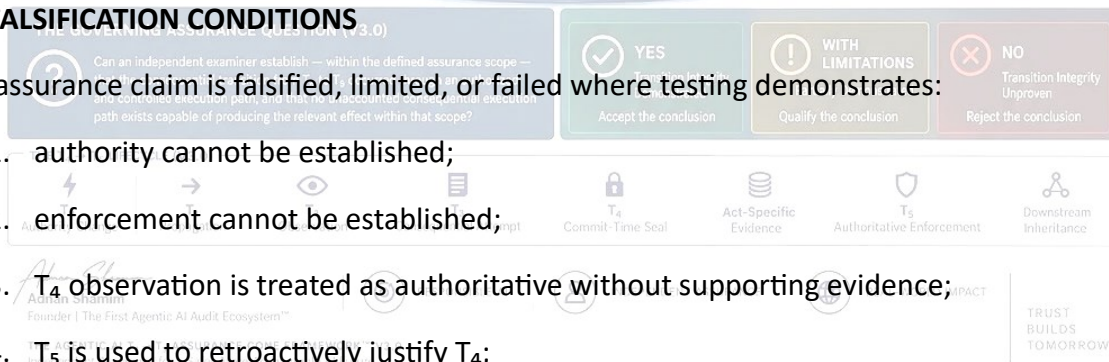
Downstream Inheritance

RESPONSIBLE AI

INDEPENDENT ASSURANCE

REAL-WORLD IMPACT

TRUST BUILDS TOMORROW



5. act-specific evidence is absent;

6. evidence is not uniquely bound to the act;

7. evidence is replayable, substitutable, or alterable;

8. an alternate path bypasses the boundary;

9. execution-path coverage cannot be demonstrated;

10. consequential execution population is undefined;

11. delegated authority expands without authorization;

Agentic AI $T_4 \rightarrow T_5$ Assurance Cone Framework™ V3.0

12. propagation exposure exceeds its predefined boundary;

13. evidence relies solely upon retrospective assertion;

14. independent reconstruction is impossible;

15. assurance conclusion exceeds demonstrated coverage;

16. a consequential path exists outside the claimed assurance boundary without explicit treatment.

20. INDEPENDENT RECONSTRUCTION TEST

An independent examiner must be able to determine:

Authority — What authority existed?

Observation — What did the actor observe?

Governing State — What authority/policy governed the act?

Enforcement — What actually controlled execution?

Path — Through which path did execution occur?

Coverage — Was that path demonstrably covered?

Commit — When did the consequential act commit?

Effect — What consequence occurred?

Lineage — What downstream actions inherited authority?

Evidence — Can the evidence be independently verified?

TRUST | ASSURE | ENABLE

FROM DECISION
TO ACCOUNTABLE EXECUTION

PEOPLE
GOVERNANCE
TECHNOLOGY
ASSURANCE
TRUST

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TURNING
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PROGRESS

SIX ASSURANCE FUNCTIONS.

1 INTENT

What was intended
and why does it matter?

- Consequential objective
- Affected resource
- Applicable constraints

2 ENFORCEMENT

What actually
controlled execution?

- Scope and duration
- Resource and action
- Policy vs actual enforcement
- Permit / deny / contain / terminate
- Control effectiveness
- No bypass paths
- Enforcement at commit point

3 PATH

Can the conclusion
be challenged?

- Parallel / asynchronous paths
- Forking / duplication / bypass
- Failure and recovery scenarios

4 EVIDENCE

Can the conclusion
be challenged?

- Forensic & Revocability
- Containment & Accountability
- Evidence Validity
- Evidence Completeness
- Coverage Must Be Demonstrated
- Unknown Coverage Is A Limitation
- No Captured Act ≠ No Act Occurred
- Assurance ≠ Assertion
- Conclusion Cannot Exceed Evidence
- Alignment ≠ External Conformity
- Framework Use ≠ Certification

5 COMMIT

When did the consequential
act commit?

- Enforcement → Commit
- Evidence Validity and provenance
- Exposure and containment
- Revocation effectiveness
- Attribution and responsibility

6 EFFECT

What consequence
occurred?

- Consequential execution
- Downstream inheritance
- Real-world impact

ENTERPRISE CONTEXT

RISK | PEOPLE | PROCESSES | TECHNOLOGY | DATA | VALUE

INTENT

Objective & Context

ENFORCEMENT

Active & Effective

PATH

Execution Path

EVIDENCE

Independent & Reconstructable

COMMIT

Consequential Execution

EFFECT

Downstream Inheritance

REAL-WORLD IMPACT

Trust Builds Tomorrow

THE $T_4 \rightarrow T_5$ ASSURANCE BOUNDARY (V3.0)

Must be established, within the defined assurance scope:

- AUTHORITY VALIDATION**
Was the relevant authority valid at the applicable commit point?
- TRANSITION CONTROL**
Were required controls active and effective?
- EXECUTION-PATH COVERAGE**
Were all consequential execution paths identified, controlled, and accounted for?
- INTERVENTION / REVOCATION EFFECTIVENESS**
If intervention or revocation occurred, did it become effective?
- EVIDENCE RECONCILIATION**
Do independent evidence sources establish a consistent transition sequence?
- BOUNDARY INTEGRITY**
Does the assurance conclusion remain within the demonstrated evidence and execution boundary?

NO ASSURANCE CLAIM MAY EXTEND BEYOND WHAT CAN BE INDEPENDENTLY DEMONSTRATED.

KEY V3.0 PRINCIPLES

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- AUTHORIZATION ≠ EXECUTION
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THE GOVERNING ASSURANCE QUESTION (V3.0)

Was that path demonstrably covered?
that the consequential transition from T_4 to T_5 occurred through an authorized and controlled execution path, and that no unaccounted consequential execution exists capable of producing the relevant effect within the scope?

YES

Transition Integrity
Demonstrated
Accept the conclusion

WITH
LIMITATIONS

Partially Demonstrated
Qualify the conclusion

NO

Transition Integrity
Unproven
Reject the conclusion

THE $T_4 \rightarrow T_5$ LIFECYCLE (V3.0)

- Authority Change
- Propagation
- Observation
- Consequential Attempt
- Commit-Time Seal
- Act-Specific Evidence
- Authoritative Enforcement
- Downstream Inheritance

Adnan Shamim

Author | The Agentic AI Assurance Framework™ V3.0

THE AGENTIC AI $T_4 \rightarrow T_5$ ASSURANCE CONE FRAMEWORK™ V3.0

Independent Assurance for a Safer, More Accountable AI Future.

RESPONSIBLE AI

INDEPENDENT ASSURANCE

REAL-WORLD IMPACT

TRUST
BUILDS
TOMORROW

Limitation — What remains outside the assurance boundary?

If material questions cannot be independently answered, the conclusion must be qualified accordingly.

21. ASSURANCE CONCLUSION MODEL

A — DEMONSTRATED

A trustworthy AI decision does not automatically prove trustworthy execution. The assurance gap is addressed by the demonstrated transition integrity. Required authority, enforcement, execution-path coverage, evidence, and reconstruction are demonstrated within the defined boundary.

B — QUALIFIED

The principal assurance condition is demonstrated, but material limitations exist.

C — NOT DEMONSTRATED

Evidence is insufficient to support the claimed assurance conclusion.

D — FAILED

Testing demonstrates violation of a required assurance condition.

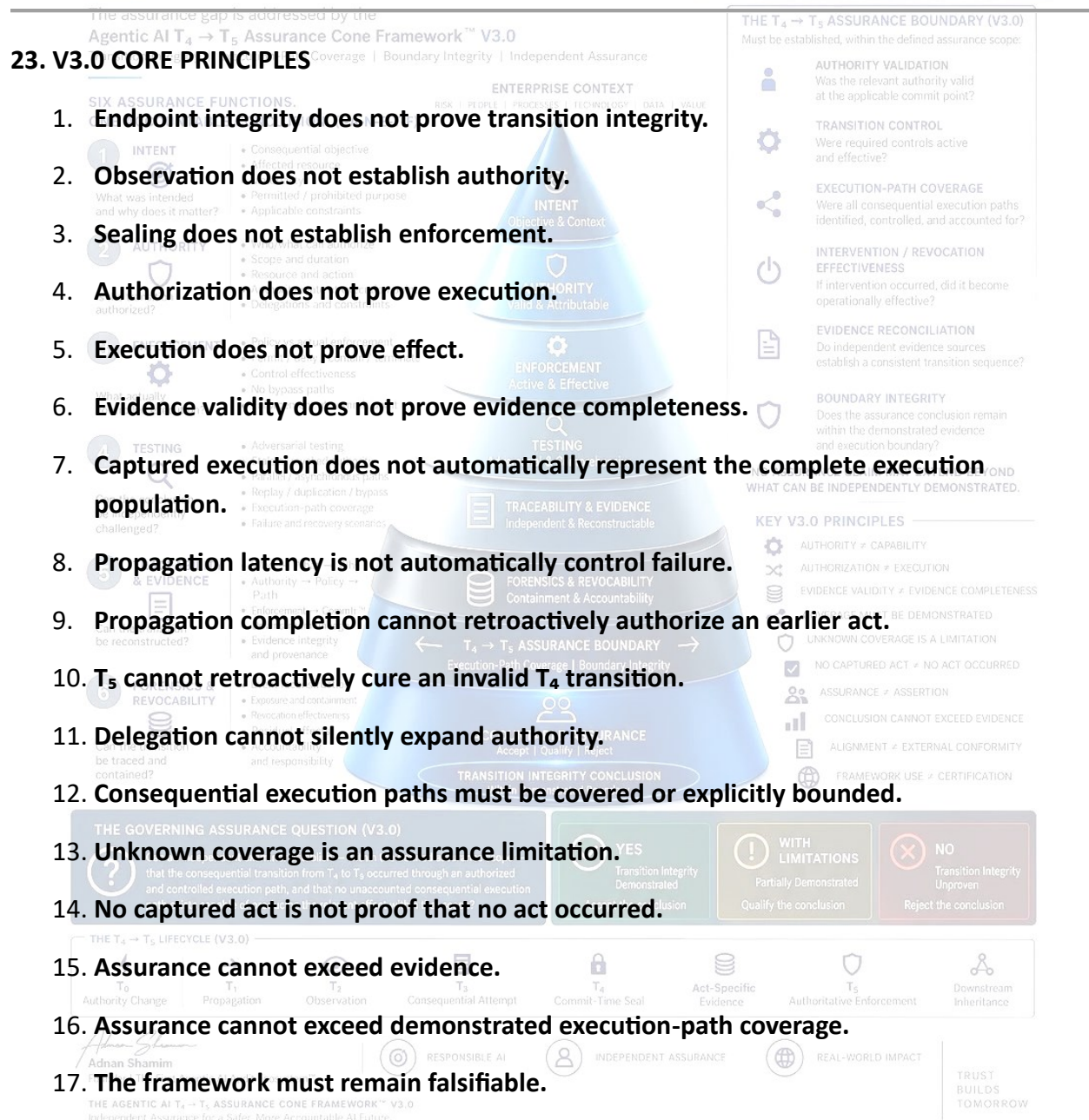
Governing Rule

The assurance classification cannot be stronger than the evidence and coverage supporting it.

22. V3.0 ASSURANCE MATRIX

Dimension	Assurance Question	Required Demonstration
Intent	What was intended?	Defined consequential objective
Authority	What authorized it?	Attributable, applicable authority
Enforcement	What controlled it?	Demonstrable enforcement
Testing	Can the claim be falsified?	Adversarial testing
Traceability	Can it be reconstructed?	Independent evidence lineage
Forensics	Can exposure/responsibility be established?	Reconstructable forensic evidence

Dimension	Assurance Question	Required Demonstration
Execution-Path Coverage	Which paths can produce consequence?	Demonstrable path coverage
Boundary Integrity	Where does the conclusion stop?	Explicit assurance boundary
External Requirements	What requirements apply?	Separate applicability/conformity assessment



18. Independent reconstruction is stronger than retrospective assertion.

19. Every material exclusion must be explicit.

20. Every material coverage failure must have a predetermined assurance consequence.

21. Framework alignment does not establish external conformity.

22. Framework application does not constitute certification.

23. An assurance conclusion does not constitute regulatory approval.

24. External requirements must be assessed independently where applicable.

24. THE V3.0 MASTER ASSURANCE QUESTION

What independently verifiable evidence establishes that every consequential execution path within the claimed assurance population is subject to the required authorization, enforcement, testing, traceability, and forensic controls—and that the resulting assurance conclusion does not extend beyond the execution paths, evidence population, applicable requirements, and boundary that can actually be demonstrated?

25. FINAL ASSURANCE POSITION

The Agentic AI $T_4 \rightarrow T_5$ Assurance Cone Framework™ V3.0 does not attempt to prove an impossible universal negative.

It establishes a disciplined assurance boundary.

T_4 establishes:

What the actor observed and sealed.

Act-specific evidence establishes:

What was bound to the consequential commit.

T_5 establishes:

What the authoritative enforcement environment established or enforced.

Execution-Path Coverage establishes:

Which consequential execution population the assurance claim can legitimately address.

Boundary Integrity establishes:

TRUST | ASSURE | ENABLE

Where the assurance conclusion must stop.

External Requirements establish:

Which additional obligations must be assessed independently.

Therefore:

Not T₄. Not T₅. The Space Between Them.

ASSURANCE ≠ ASSERTION does not automatically prove trustworthy execution.

ASSURANCE = DEMONSTRATED CONTROL + DEMONSTRATED COVERAGE + INDEPENDENTLY VERIFIABLE EVIDENCE + EXPLICIT BOUNDARY

THE GOVERNING V3.0 RULE

ONE ACCOUNTABLE CONCLUSION. (I-A-N-E-T-F)

IF THE EVIDENCE BOUNDARY IS NARROWER THAN THE CONSEQUENTIAL EXECUTION BOUNDARY, THE ASSURANCE CONCLUSION MUST BE NARROWER TOO.

THE CLAIMS DISCIPLINE RULE

NO ASSURANCE CLAIM MAY EXTEND BEYOND THE AUTHORITY, ENFORCEMENT, EXECUTION-PATH COVERAGE, EVIDENCE, AND BOUNDARY THAT CAN BE INDEPENDENTLY DEMONSTRATED.

THE INDEPENDENCE RULE

V3.0 MAY COMPLEMENT OTHER REQUIREMENTS, BUT IT DOES NOT SUBSTITUTE FOR THEM, AND ITS APPLICATION DOES NOT BY ITSELF ESTABLISH EXTERNAL CONFORMITY, CERTIFICATION, OR REGULATORY COMPLIANCE.

This preserves the framework as an independent, technology-neutral assurance architecture without attaching it to, naming, or implying endorsement by any external institution or standard.

Attribution & Intellectual Lineage:

The Agentic AI T₄→T₅ Assurance Cone Framework™ acknowledges Randy Feldman's documented August 26, 2026 articulation of the T₄ as observed-and-sealed against T₅ as authoritatively enforced distinction, his August 31, 2026 contribution regarding the evidence validity versus evidence completeness scope line, and discussions with Jim Rosenberger that helped refine the thinking informing this work.

How Do We Prove Transition Integrity in Agentic AI?

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The assurance gap is addressed by the
Agentic AI T₄ → T₅ Assurance Cone Framework™ V3.0

Transition Integrity | Execution-Path Coverage | Boundary Integrity | Independent Assurance

SIX ASSURANCE FUNCTIONS.

ONE ACCOUNTABLE CONCLUSION. (I-A-N-E-T-F)

- 1 INTENT**
What was intended and why does it matter?
 - Consequential objective
 - Affected resource
 - Materiality
 - Permitted / prohibited purpose
 - Applicable constraints
- 2 AUTHORITY**
Is the transition authorized?
 - Who/what can authorize
 - Scope and duration
 - Resource and action
 - Authority state at execution point
 - Delegations and constraints
- 3 ENFORCEMENT**
What actually controlled execution?
 - Policy vs actual enforcement
 - Permit / deny / contain / terminate
 - Control effectiveness
 - No bypass paths
 - Enforcement at commit point
- 4 TESTING**
Can the conclusion be independently challenged?
 - Adversarial testing
 - Stale or revoked authority
 - Parallel / asynchronous paths
 - Replay / duplication / bypass
 - Execution-path coverage
 - Failure and recovery scenarios
- 5 TRACEABILITY & EVIDENCE**
Can the transition be reconstructed?
 - Who → What → Why
 - Authority → Policy → Path
 - Enforcement → Commit™
 - Downstream lineage
 - Evidence integrity and provenance
- 6 FORENSICS & REVOCABILITY**
Can the transition be traced and contained?
 - Event reconstruction
 - Exposure and containment
 - Revocation effectiveness
 - Residual effects
 - Accountability and responsibility

ENTERPRISE CONTEXT
RISK | PEOPLE | PROCESSES | TECHNOLOGY | DATA | VALUE



THE T₄ → T₅ ASSURANCE BOUNDARY (V3.0)

Must be established, within the defined assurance scope:

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Was the relevant authority valid at the applicable commit point?
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- EXECUTION-PATH COVERAGE**
Were all consequential execution paths identified, controlled, and accounted for?
- INTERVENTION / REVOCATION EFFECTIVENESS**
If intervention occurred, did it become operationally effective?
- EVIDENCE RECONCILIATION**
Do independent evidence sources establish a consistent transition sequence?
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- FRAMEWORK USE ≠ CERTIFICATION

THE GOVERNING ASSURANCE QUESTION (V3.0)

Can an independent examiner establish — within the defined assurance scope — that the consequential transition from T₄ to T₅ occurred through an authorized and controlled execution path, and that no unaccounted consequential execution path exists capable of producing the relevant effect within that scope?

YES
Transition Integrity Demonstrated
Accept the conclusion

WITH LIMITATIONS
Partially Demonstrated
Qualify the conclusion

NO
Transition Integrity Unproven
Reject the conclusion

THE T₄ → T₅ LIFECYCLE (V3.0)



Adnan Shamim
Founder | The First Agentic AI Audit Ecosystem™



RESPONSIBLE AI



INDEPENDENT ASSURANCE



REAL-WORLD IMPACT

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